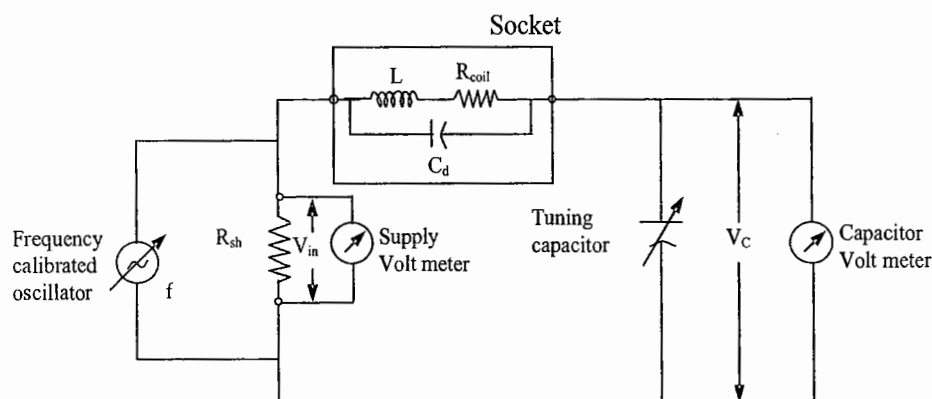


Q - Meter

- The Q-meter is an instrument designed for the measurement of Q-factor of the coil as well as for the measurement of electrical properties of coils like Self inductance, Self capacitance, Resistance. This instrument operates on the principle of series resonance i.e. at resonate condition of an ac series circuit voltage across the capacitor is equal to Q-times of the applied voltage to the circuit. If the voltage applied across the circuit is kept-constant then voltmeter connected across the capacitor can be calibrated to indicate Q directly.
- Circuit diagram of "Direct connection" of a Q-meter is shown in figure. A wide-range os-

cillator with frequency range from 50 kHz to 50 MHz is used as a power supply to the circuit. The output of the oscillator is shorted by a low-value resistance, R_{sh} usually of the order of 0.02 ohm. So it introduces almost no resistance into the oscillatory circuit and represents a voltage source with a very small or of almost negligible internal resistance. The voltage across the low-value shunt resistance R_{sh} , V is measured by a thermocouple meter and the voltage across the capacitor, V_c is measured by an electronic voltmeter.



- For carrying out the measurement, the unknown coil is connected to the test terminals of the instrument, and the circuit is tuned to resonance either by varying the frequency of the oscillator or by varying the resonating capacitor C . Readings of voltages across capacitor C and shunt resistance R_{sh} are obtained and Q-factor of the coil is determined as follows:
- By definition Q-factor of the coil,
 $Q = X_L / R$
 And when the circuit is under resonance condition: $X_L = X_C$
 i.e., $Q = X_C / R$

$$\begin{aligned}
 V_c &= IX_c \\
 &= (V_{in}/Z) X_c \\
 &= (V_{in}/R) X_c \quad (\text{because } Z=R \text{ at resonance}) \\
 &= (X_c/R) V_{in} \text{ or } (X_L/R) V_{in} \\
 &= Q V_{in}
 \end{aligned}$$

Where Q is Known as Voltage-magnification factor.

$$\begin{aligned}
 Q &= V_c / V_{in} \\
 &= \text{Q-Capacitor Voltmeter /Supply Voltmeter reading}
 \end{aligned}$$

- This relation($Q = V_c/V_{in}$) is used to calibrate the voltscale of capacitor voltmeter into Q-scale



- This Q-factor is called the circuit Q because this measurement includes distributed capacitance of coil, shunt resistor R_{sh} . So, the calculated Q-factor of the coil will be somewhat less than the Actual Q-factor.
- There are two more connections of Q meter namely series connection and shunt connection as given below.

At resonance,

$$\omega L = \frac{1}{\omega C_1} \text{ and } Q_1 = \frac{\omega L}{R} = \frac{1}{\omega C_1 R}$$

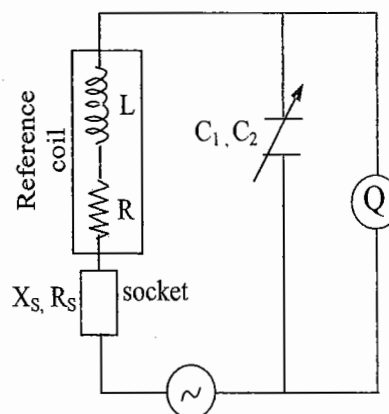
$$X_s = \frac{C_1 - C_2}{\omega C_1 C_2}$$

$$R_s = \frac{Q_1 C_1 - Q_2 C_2}{\omega C_1 C_2 Q_1 Q_2}$$

$$Q_x = \frac{X_s}{R_s} = \frac{(C_1 - C_2) Q_1 Q_2}{Q_1 C_1 - Q_2 C_2}$$

i) Series Connection:

In series connection method the unknown test impedance is connected in series with the Reference coil. This method is employed for measurement of low test impedance i.e., low value resistors, small coils and large capacitors.



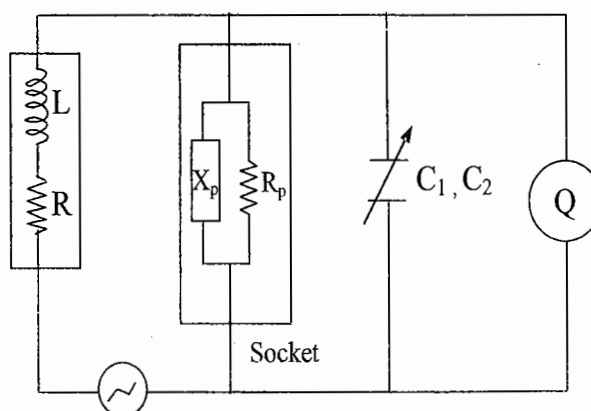
ii) Parallel Connection:

In parallel connection, the unknown test impedance is connected in parallel to the tuning capacitor. This method is employed for measurement of high impedance i.e., high value resistors, certain inductors and small capacitors.

$$\frac{1}{R_p} = \frac{\omega C_1}{Q_2} = \frac{1}{R Q_1}$$

$$X_p = \frac{1}{\omega (C_2 - C_1)}$$

$$Q_x = \frac{R_p}{X_p} = \frac{(C_1 - C_2) Q_1 Q_2}{Q_1 C_1 - Q_2 C_2}$$



The main error in the measurement of Q is due to distributed or stray capacitance of the circuitry. To check for this, the Q is measured at two frequencies f_1 and $2f_1$. It should be same if not,

$$C_d = \frac{C_1 - 4C_2}{3}$$